Cumulative A05 profile

Transcript copy constructor 351.96 CPU s 10.90% exclusive CPU

Select conservative ownership change before compact rollback redesign

A05 recursive stitching

Transcript passed by value

Include edge copy then mutate

Exclude edge copy then only forward

Terminal path mutate and finalize copy

same search tree and branch order

A06 accepted in blackstar.2

Recursive input is const reference

Branch type

include

exclude

terminal

Include make private copy then mutate

Exclude forward same state without a copy

Terminal make private copy then finalize

Exact public evidence at 96 threads

Copy CPU 351.96 s to 225.35 s 35.97% reduction

Five-pair uncompressed median 57.39 s to 56.11 s 2.29% gain

Bootstrap interval 1.52% to 3.79% 5 of 5 outputs passed

Gzip median gain 4.59% canonical BAM passed ASan and UBSan passed peak RSS unchanged